Bhalaji Yadav Kantepalle

Ph.D. Student, Researcher, and Chemical Engineer

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RESEARCH PROFILE

Integrative Life Sciences Ph.D. student and chemical engineer studying stimuli-responsive soft and liquid-crystalline materials through mechanics, quantitative imaging, rheology, and structure-sensitive characterization. Develops reproducible experimental and computational workflows that connect processing, structure, material response, and engineering decisions.

EDUCATION

Ph.D. Student, Integrative Life Sciences — Aug 2026–Present
Virginia Commonwealth University — *Richmond, Virginia*

M.S., Chemical and Life Science Engineering — Aug 2026
Virginia Commonwealth University — *Richmond, Virginia*

- GPA: 3.62
- Thesis: Multi-Criteria Selection of Adhesives for Wearable Textiles
- Advisor: Christina Tang, Ph.D.

B.Tech., Chemical Engineering — Jun 2022
Amrita Vishwa Vidyapeetham — *Coimbatore, India*

RESEARCH AND TEACHING EXPERIENCE

Graduate Teaching Assistant — Aug 2026–Present
Virginia Commonwealth University — *Richmond, Virginia*

- Support undergraduate instruction while pursuing doctoral training in the Integrative Life Sciences Ph.D. program.

Graduate Researcher, Quantitative Metrology — Sep 2024–Aug 2026
VCU Soft Functional Materials Lab | Advisor: Christina Tang, Ph.D. — *Richmond, Virginia*

- Developed mechanics-consistent adhesion and fracture-characterization methods for soft, stretchable, and textile-supported interfaces.
- Engineered reproducible Python and OpenCV workflows that synchronized force-displacement data, deformation tracking, and optical colorimetry.
- Guided two ongoing Summer 2026 student projects in liquid-crystal temperature mapping and hyperspectral detection of plastics in soil mixtures; provided Spring 2026 technical support in NMR characterization, strain-to-color analysis, scientific figures, posters, and grant drafting.
- Contributed technical writing and regulatory analysis to a phased $100,000 Commonwealth Cyber Initiative proposal; the $30,000 prototype and proof-of-concept phase was initially funded, with later funding contingent on satisfactory progress.

PEER-REVIEWED PUBLICATIONS

1. Kantepalle, B. Y.; Epitawala Arachchige, U.; Joung, D.; Tang, C. Multi-Criteria Selection of Adhesives for Wearable Textiles. Polymers 2026, 18, 1504. https://doi.org/10.3390/polym18121504
2. Caloian, I.; Trapp, J.; Kantepalle, B. Y.; Latimer, P.; Lawton, T. J.; Tang, C. Mechanical Properties of Dual-Layer Electrospun Fiber Mats. Polymers 2025, 17, 1777. https://doi.org/10.3390/polym17131777

RESEARCH SOFTWARE

Peel Trace Evaluation for Soft Substrates — v1.4.0-rc15
Bhalaji Yadav Kantepalle and Christina Tang
Open-source Jupyter and Colab workflow for protocol-defined analysis of force-displacement peel traces from soft substrates. Repository | Archived release

PRESENTATIONS, SEMINAR, AND DEFENSE

1. **M.S. Thesis Defense.** B. Y. Kantepalle, "Multi-Criteria Selection of Adhesives for Wearable Textiles," Virginia Commonwealth University, Richmond, Virginia. — Jul 1, 2026

2. **Poster Presentation.** B. Y. Kantepalle, U. E. Arachchige, D. Joung, and C. Tang, "Multi-Criteria Selection of Adhesives for Wearable Textiles," PMDC 4th Annual Research Symposium, Virginia Commonwealth University, Richmond, Virginia. — Jun 12, 2026
3. **Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," 29th VCU Graduate Student Research Symposium, Richmond, Virginia. — Apr 22, 2026
4. **Departmental Research Seminar.** B. Y. Kantepalle, "Beyond Peel Strength: Stability-Driven Adhesion Metrics for Soft Wearable Interfaces," Chemical and Life Science Engineering Research Day, Virginia Commonwealth University, Richmond, Virginia. — Apr 15, 2026
5. **Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. — Mar 25, 2026
6. **Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," ACS Fall 2025, Washington, DC. — Aug 20, 2025
7. **Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," 28th VCU Graduate Student Research Symposium, Richmond, Virginia. — Apr 23, 2025
8. **Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. — Mar 26, 2025

TEACHING, REVIEW, AND APPLIED INNOVATION

Classroom Instructor, Undergraduate Material Balance — Nov 11–12, 2025
Virginia Commonwealth University

Judge, Student Poster Session — Mar 29, 2026
AIChE Atlantic Regional Conference, Virginia Commonwealth University

Judge, Chem-E-Car Poster Competition — Mar 28, 2026
AIChE Mid-Atlantic Regional Conference, Virginia Commonwealth University

Prototype Demonstrator, Feedback Friday — Mar 20, 2026
Shelfie Program, VCU da Vinci Center

Student Volunteer — Feb 2020
Anokha, Amrita University Techfest

INDUSTRY EXPERIENCE

Technical Project Manager — May 2023–May 2024
Kreative Organics Pvt. Ltd. — *Hyderabad, India*

- Coordinated quality, engineering, and external stakeholders during a regulated digital-transformation initiative.
- Translated process requirements into validation documentation, traceable test protocols, and data-integrity controls.
- Standardized chemical nomenclature and analytical workflows to support more consistent technical and business decisions.

Technical Project Manager Intern — Oct 2022–Apr 2023
Kreative Organics Pvt. Ltd. — *Hyderabad, India*

- Supported digital-transformation, chemical-data standardization, and technical-documentation workflows in regulated manufacturing.

HONORS AND PROFESSIONAL DEVELOPMENT

- Graduate Assistantship Award, Virginia Commonwealth University, Fall 2025
- Selected Participant, 1st National Neutron Scattering School, Oak Ridge National Laboratory, Sep 2025
- Design Thinking, Prototyping, and Pitching & Storytelling credentials, VCU da Vinci Center, Mar 2026

TECHNICAL SKILLS

Materials and Testing: Adhesion testing, T-peel, fracture mechanics, tensile testing, rheology, viscoelasticity, soft interfaces, wearable textiles

Computational: Python, OpenCV, NumPy, Pandas, SciPy, Matplotlib, image processing, CIE L*a*b* colorimetry, automated analysis

Instrumentation: Universal testing machine, optical microscopy, UV-Vis spectroscopy, Abbe refractometry, 3D printing

Compliance and Process: Validation documentation, data integrity, cGMP, V-model, RCA, CAPA, DOE, Lean manufacturing